

MSI 5102 Type 2-Projects

1. Handwritten Digit Recognition

Goal: Classify handwritten digits.

Dataset

- MNIST dataset

Tasks

- Visualize sample images
- Train models:
 - k-Nearest Neighbors
 - Logistic Regression
 - Neural Network (optional)
- Use **t-SNE** or **PCA** to visualize the digits in 2D.

Learning Objectives

- Image data representation
- Multiclass classification
- Dimensionality reduction